A. Clinical ECG sources

PTB-XL



20,487 ECGs

Georgia



8,302 ECGs

Chapman



8,634 ECGs

Ningbo



28,956 ECGs

Harmonised label space

13 diagnostic classes retained across all sources

Leave-one-source-out evaluation

Fold 1	Fold 2	Fold 3	Fold 4
test: PTB-XL	test: Georgia	test: Chapman	test: Ningbo

train on three sources, test on the fourth

B. Reference models and deployment-oriented scaling

10 reference models (unconstrained)

published topology, float32, no device budget

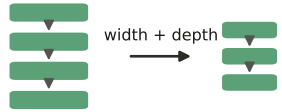
Two scaling strategies for MCU deployment

Uniform width scaling



thinner, same depth

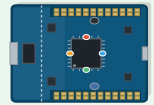
Joint width-depth scaling



thinner and shallower

Training, quantisation, MCU deployment

Train (FP32) → **Quantise** (INT8) → **Deploy on MCU**



measured on STM32F411VET6:
Flash, peak SRAM, latency, macro-AP

C. Evaluation outputs

RQ1. Cross-source generalization

- external discrimination, four held-out sources
- class-specific AP and paired contrasts

RQ2. Scaling and MCU efficiency

- scaling strategy at matched budgets
- quantisation, Flash, SRAM and latency

RQ3. Performance-resource trade-offs

- macro-AP against measured latency and SRAM
- non-dominated set, both methods pooled